The Fundamental Theorem of Arithmetic Proof

Kian Farr

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1 The Fundamental Theorem of Arithmetic Part 1 (Existence)

Proof. **Base Case:** Let $n = 2$. Then let $S = 2$, and $f(2) = 1$. Then $\prod_{i \in S} i^{f(i)} = 2^1 = 2$. Thus, the base case holds.

Inductive Step (Strong Induction): Suppose that $P(m)$ holds for all $1 < m < n$. Let p be the smallest factor of n other than 1. It can be shown that such a factor must exist and is prime. There exists some $1 < m < n$ such that $p \cdot m = n$ (if n is not prime, otherwise if n is prime, the claim is trivial). Let S be the set satisfying $P(m)$ and f be the function satisfying $P(m)$. If $p \in S$, then define $g : S \to \mathbb{N}$ by

$$g(x) = \begin{cases} f(x) + 1 & \text{if } x = p \\ f(x) & \text{otherwise.} \end{cases}$$

It can be shown that S and g satisfy $P(n)$. If $p \notin S$, then define $g : S \cup \{p\} \to \mathbb{N}$ by

$$g(x) = \begin{cases} f(x) & \text{if } x \in S \\ 1 & \text{if } x = p. \end{cases}$$

It can be shown that S and g satisfy $P(n)$. Thus, the inductive step holds. $\square$

2 The Fundamental Theorem of Arithmetic Part 2 (Uniqueness)

Proof. Let $p \in S_1$. Then

$$p \mid \prod_{q \in S_2} q^{g(q)}.$$

Thus, $p \mid q^{g(q)}$ for some $q \in S_2$. This implies that $p \mid q$ and that $p = q$, since p and q are both prime. Thus, $S_1 \subseteq S_2$. A symmetric argument shows that

$S_2 \subseteq S_1$. Now suppose that $f(p) > g(p)$ for some $p \in S_1$. Then

$$p \mid \frac{\displaystyle\prod_{q \in S_1} q^{f(q)}}{p^{g(p)}},$$

but

$$p \nmid \frac{\displaystyle\prod_{q \in S_2} q^{g(q)}}{p^{g(p)}}$$

(because of the uniqueness of the set of prime factors of $\frac{\prod_{q \in S_2} q^{g(q)}}{p^{g(p)}}$), which is a contradiction. Thus, $f(p) \leq g(p)$. A symmetric argument shows that $f(p) \geq g(p)$. Thus, $f(p) = g(p)$ for all $p \in S_1$. $\square$